

Technical Report for Data Visualization

Investigating the Shift in European Car Registrations

Gianluca Rigatti - ID 265303

Giuseppe Screnci - ID 263989

1 Introduction

The European car market has been significantly transformed in the last two decades to embrace the transition towards more sustainable forms of road transportation. As observing the increasing registrations of electrified vehicles does not provide an accurate picture of the ongoing market transition, we wish to understand the true market offerings by power train type while accounting for the simultaneous decrease of offerings in ICE power trains.

This document presents the datasets selected for the development of an interactive data visualization solution aimed at exploring the evolution of new passenger car registrations in Europe. The European Environment Agency (EEA) dataset serves as the primary data source, providing detailed technical specifications of newly registered vehicle models. Complementary registration datasets published by Eurostat and the United Nations Economic Commission for Europe (UNECE) are investigated as potential validation sources to evaluate the consistency of the data resulting from the preprocessing steps.

In this report, we assess the quality and suitability of the available datasets, identify the most appropriate sources for the subsequent analyses, and illustrate the preprocessing steps required to construct a consistent dataset for the proposed visualization project. An exploratory data analysis (EDA) is performed on the datasets following the general workflow proposed in the literature by *Alshammari* [2] and *Agarwal et al.* [1]. Our analysis is structured into the following stages:

- *Dataset description.* We describe the origin, structure, temporal and geographical coverage and metadata of each dataset to obtain a general overview.
- *Data inspection, cleaning and preprocessing.* Each dataset's attribute is inspected to identify missing values, invalid observations (e.g. NaN, NULL and Infinity), duplicate records, outliers and resolve general inconsistencies among datasets. Irrelevant attributes are also removed, to obtain a format suitable for subsequent analyses. Given the size of the EEA dataset, it is also converted to a more efficient format to reduce its memory requirements and make it easier to process. Finally, the registration counts derived from the preprocessed EEA dataset are validated against the aggregated statistics from the complementary datasets.
- *Univariate, Bivariate and Multivariate analysis.* Relationships within the data are investigated to uncover patterns and trends. In the Univariate analysis step, summary statistics are computed to provide insights into the central tendency, spread and distribution of the data. In the Bivariate and Multivariate analysis steps, correlations between variables are explored in order to understand how changes in one variable relate to changes in other variables. In this latter stage heatmaps are used as they allow for visual recognition of patterns and trends.
- *Time series analysis.* Given the time-based nature of vehicle registrations, time series analysis is employed to analyse and forecast short-term direction of vehicle registrations.
- *Feature selection.* Relevant features are selected according to the requirements of the two normalisation approaches described in the project proposal. For the attribute-based method, a set of attributes from the cleaned EEA dataset is selected to provide a representation of user choice. For the Autoencoder-based method, vehicle specifications are used as input variables for training the neural network, obtaining a latent space that captures the utility offered by different fuel type options.

To support our analysis, we make use of the `fg-data-profiling` library [4], which enables automatic generation of comprehensive profiling reports. These reports provide a baseline overview of the datasets and facilitate the identification of potential data quality issues before subsequent analysis and visualizations.

2 Datasets description

2.1 Eurostat’s New passenger cars by type of motor energy and engine size

The *New passenger cars by type of motor energy and engine size or road_eqr_carmot* dataset, published by Eurostat, contains annual statistics on newly registered passenger cars across European countries. Registrations are disaggregated by country, year, type of motor energy and engine size class, allowing the analysis of the evolution of vehicle power train choices over time. The structure of the dataset is presented in **Tab. 1**. The dataset is stored in the CSV format.

Table 1: Description of the attributes in the Eurostat *New passenger cars by type of motor energy and engine size* dataset.

Attribute	Type	Description
STRUCTURE	Categorical	
STRUCTURE_ID	Categorical	
STRUCTURE_NAME	Categorical	
freq	Categorical	Code representing the frequency of observations.
Time frequency	Categorical	Description of the observation frequency 'freq'.
unit	Categorical	Code representing the measurement unit.
Unit of measure	Categorical	Description of the measurement unit 'unit'.
mot_nrg	Categorical	Code representing the motor energy category.
Motor energy	Categorical	Description of the motor energy category 'mot_nrg'.
engine	Categorical	Code representing the engine capacity category.
Engine capacity of vehicle	Categorical	Description of the engine capacity category 'engine'.
geo	Categorical	ISO2 country code of the reporting country.
Geopolitical entity (reporting)	Categorical	Description of the reporting country 'geo'.
TIME_PERIOD	Numerical	Year of the observation.
Time	NaN	
OBS_VALUE	Numerical	Number of new passenger cars registrations.
Observation value	NaN	
OBS_FLAG	Categorical	Flag indicating special conditions of the observation.
Observation status (Flag) V2 structure	Categorical	Description of the flag 'OBS_FLAG'. (d) is definition differs (see metadata), (e) is estimated, (b) is break in time series, (i) is value imputed by Eurostat or other receiving agencies, (u) is low reliability, (p) is provisional.
CONF_STATUS	Categorical	Confidentiality status of the observation.
Confidentiality status (flag)	Categorical	Description of the flag 'CONF_STATUS'. (C) is confidential (value hidden to protect identity).

The dataset covers annual observations over the period from **1970** to **2024**. From a geographical perspective, the dataset includes **42** reporting countries. Although the dataset primarily focuses on European countries, the geographical coverage is not restricted to European Union Member States as it also includes EFTA countries and other participating countries.

2.2 Eurostat’s New passenger cars by type of motor energy

The *New passenger cars by type of motor energy or road_eqr_carpda* dataset, published by Eurostat, contains annual statistics on newly registered passenger cars across European countries. Registrations are disaggregated by country, year and type of motor energy, allowing the analysis of the evolution of vehicle power train choices over time. The structure of the dataset is presented in **Tab. 2**. The dataset is stored in the CSV format.

Table 2: Description of the attributes in the Eurostat *New passenger cars by type of motor energy* dataset.

Attribute	Type	Description
STRUCTURE	Categorical	
STRUCTURE_ID	Categorical	
STRUCTURE_NAME	Categorical	
freq	Categorical	Code representing the frequency of observations.
Time frequency	Categorical	Description of the observation frequency 'freq'.
unit	Categorical	Code representing the measurement unit.
Unit of measure	Categorical	Description of the measurement unit 'unit'.
mot_nrg	Categorical	Code representing the motor energy category.
Motor energy	Categorical	Description of the motor energy category 'mot_nrg'.
geo	Categorical	ISO2 country code of the reporting country.
Geopolitical entity (reporting)	Categorical	Description of the reporting country 'geo'.
TIME_PERIOD	Numerical	Year of the observation.
Time	NaN	
OBS_VALUE	Numerical	Number of new passenger cars registrations.
Observation value	NaN	
OBS_FLAG	Categorical	Flag indicating special conditions of the observation.
Observation status (Flag) V2 structure	Categorical	Description of the flag 'OBS_FLAG'. (i) is value imputed by Eurostat or other receiving agencies, (e) is estimated, (b) is break in time series, (m) is missing value; data cannot exist.
CONF_STATUS	NaN	
Confidentiality status (flag)	NaN	

The dataset covers annual observations over the period from **2013** to **2024**. From a geographical perspective, the dataset includes **42** reporting countries. Although the dataset primarily focuses on European countries, the geographical coverage is not restricted to European Union Member States as it also includes EFTA countries and other participating countries.

2.3 UNECE’s New passenger car registrations by fuel type (detailed)

The *New passenger car registrations by fuel type (detailed)* dataset, published by the United Nations Economic Commission for Europe (UNECE), contains annual statistics on newly registered passenger cars reported by participating countries.

The original dataset is provided in a MultiIndex structure rather than a tabular structure. It consists of a hierarchical index composed of fuel type, country and year, with registration values as data. The original structure of the dataset is presented in **Tab. 3**, while additional metadata for specific countries observations is available on the dataset’s website. The dataset is stored in the CSV format.

Table 3: Description of the attributes in the original UNECE *New passenger car registrations by fuel type (detailed)* dataset.

Attribute	Type	Description
Fuel Type	Categorical	Category of motor energy and engine capacity.
Country	Categorical	Reporting country.
Year	Categorical	Year of the observation.
New passenger car registrations by fuel type (detailed)	Numerical	Number of newly registered passenger cars. Missing observations are represented by the symbol "...".

Given that the raw structure is not directly suitable for subsequent analyses, a preprocessing step is performed to obtain a long format representation of the data. Furthermore, the original hierarchical fuel classification is decomposed into separate, explicit attributes. To facilitate integration with the Eurostat datasets, the attribute names are harmonized and a **geo** attribute containing the ISO2 country codes is introduced. In the resulting dataset, each row corresponds to a single observation identified by country, year, motor energy category and engine category. The structure of the preprocessed dataset is presented in **Tab. 4**.

Table 4: Description of the attributes in the preprocessed UNECE dataset.

Attribute	Type	Description
Motor energy	Categorical	Description of the motor energy category.
Engine capacity of vehicle	Categorical	Description of the engine capacity category.
geo	Categorical	ISO2 country code of the reporting country.
Geopolitical entity (reporting)	Categorical	Description of the reporting country 'geo'.
TIME_PERIOD	Numerical	Year of the observation.
OBS_VALUE	Numerical	Number of new passenger cars registrations.

The UNECE’s dataset covers annual observations from **1993** to **2023**. Compared to Eurostat, the geographical coverage is broader, including **56** reporting countries primarily in Europe and Asia.

2.4 EEA’s Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631

The *Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631* dataset, published by the European Environment Agency (EEA), contains the CO₂ emissions for newly registered passenger cars in a year range from **2013** to **2023** (a larger range is available on the website, though due to dataset size constraints and the provisional nature of some of the more recent data points, this specific range was chosen a priori). This dataset was selected for this kind of analysis as we required more granular data than the one offered by the aforementioned datasets, in particular, this dataset provides us with information at the level of every vehicle model and variant registered in every EU member state. This additional information is required to characterise the available vehicle market offerings. The dataset is provided in CSV format and has a size of approximately 17GB. A description of the dataset’s attributes is provided in **Tab. 5**.

3 Data inspection, cleaning and preprocessing

Consistent with the scope of this study, only observations referring to the **EU27_2020** member states are retained. The list of member states was sourced from the official Eurostat *Member States EU27 (from 2020)* classification [6]. Records not associated with these member states are removed whenever present.

3.1 Eurostat’s New passenger cars by type of motor energy and engine size

As an initial inspection step, the cardinality of each attribute was examined by computing the number of distinct values. Several attributes were found to contain either a single unique value throughout the dataset or no valid observations (NaN values). Specifically, the attributes **STRUCTURE**, **STRUCTURE_ID**, **STRUCTURE_NAME**, **freq**, **Time frequency**, **unit**, **Unit of measure**, **Time**, and **Observation value** were identified and, as they do not provide any analytical value, can be discarded and not considered for subsequent analyses.

Furthermore, Eurostat provides several pairs of attributes consisting of a code identifier and its corresponding description (e.g., **mot_nrg** and **Motor energy**). Since both encode the same information, descriptive attributes were retained when they improved readability, whereas coded attributes were preserved when required for dataset integration with other sources (e.g., **geo**). The remaining attributes were inspected to assess their characteristics and data quality. The results of this inspection are summarised in **Tab. 6**.

No duplicated rows were identified in the dataset. The only NaN values recorded correspond to 16 observations for which **OBS_VALUE** is unavailable due to confidentiality (**CONF_STATUS** = 'C'), all **before 2009**.

Table 5: Description of the attributes in the EEA’s *Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631* dataset. [Source]

Attribute	Type	Description
ID	Numerical	Identification number.
MS	Categorical	Member state.
Mp	Categorical	Manufacturer pooling.
VFN	Numerical	Vehicle family identification number.
Mh	Textual	Manufacturer name EU standard denomination .
Man	Textual	Manufacturer name OEM declaration.
MMS	Textual	Manufacturer name MS registry denomination .
TAN	Categorical	Type approval number.
T	Categorical	Type.
Va	Categorical	Variant.
Ve	Categorical	Version.
Mk	Categorical	Make.
Cn	Textual	Commercial name.
Ct	Categorical	Category of the vehicle type approved.
Cr	Categorical	Category of the vehicle registered.
M (kg)	Numerical	Mass in running order Completed/complete vehicle .
Mt	Numerical	WLTP test mass.
Enedc (g/km)	Numerical	Specific CO2 Emissions (NEDC).
Ewltp (g/km)	Numerical	Specific CO2 Emissions (WLTP).
W (mm)	Numerical	Wheel Base.
At1 (mm)	Numerical	Axle width steering axle.
At2 (mm)	Numerical	Axle width other axle.
Ft	Categorical	Fuel type.
Fm	Categorical	Fuel mode.
Ec (cm3)	Numerical	Engine capacity.
Ep (KW)	Numerical	Engine power.
Z (Wh/km)	Numerical	Electric energy consumption.
IT	Categorical	Innovative technology or group of innovative technologies.
Ernedc (g/km)	Numerical	Emissions reduction through innovative technologies.
Erwltp (g/km)	Numerical	Emissions reduction through innovative technologies (WLTP).
De	Numerical	Deviation factor.
Vf	Numerical	Verification factor.
R	Numerical	Total new registrations.
Year	Numerical	Reporting year.
Status	Categorical	P = Provisional data, F = Final data.
Version_file	Textual	Internal versioning of deliverables.
E (g/km)	Numerical	Specific CO2 Emission. Deprecated value, only relevant for data until 2016.
Er (g/km)	Numerical	Emissions reduction through innovative technologies. Deprecated value, only relevant for data until 2016.
Zr	Numerical	Electric range.
Dr	Date	Registration date.
Fc	Numerical	Fuel consumption.

Table 6: Attribute inspection of the Eurostat *New passenger cars by type of motor energy and engine size* dataset after removal of unnecessary attributes.

Attribute	Dtype	Value domain / range	Missing (%)	Notes
Motor energy	object	Diesel, Petrol, Total	0/6422 (0.0%)	
Engine capacity of vehicle	object	From 1 400 to 1 999 cm³, 2 000 cm³ or over, Less than 1 400 cm³, Total	0/6422 (0.0%)	
geo	object	ISO2 codes	0/6422 (0.0%)	39 EU, 3 Asia and EU27_2020.
Geopolitical entity (reporting)	object		0/6422 (0.0%)	Description for 'geo'.
TIME_PERIOD	int64	1970-2024	0/6422 (0.0%)	Continuous from 1979 to 2024.
OBS_VALUE	float64	0.0-13,235,768.0	0/6422 (0.0%)	42/6422 zeros (0.7%)
OBS_FLAG	object	d, e, b, i, u, p	5838/6422 (90.9%)	
CONF_STATUS	object	C	6406/6422 (99.8%)	

An inspection of zero-valued observations was then performed to determine whether they represented invalid measurements or meaningful cases of no registrations. As shown in **Fig. 1a**, zeros are distributed across multiple years but concentrated in a limited number of countries. Moreover, **Fig. 1b** shows that countries reporting zero registrations generally correspond to those with a lower total number of car registrations. Therefore, these values are considered valid observations, as they represent cases where no vehicles matching a specific combination of fuel type and engine capacity category were registered during a given year.

Second, the observation flags were inspected to assess potential issues. The most relevant flags for the analysis were identified as **b** (break in time series) and **d** (definition differs), as they may affect the temporal consistency of the comparisons. The distribution of these flags was analysed over time: **b** flagged observations were concentrated **before 2014**, whereas **d** flagged observations were all **before 2013**.

In light of these findings, the period from **2014 onwards** is considered the most suitable time window for subsequent analyses, as it reduces the influence of methodological inconsistencies and differences in reporting practices while still providing a sufficiently long period to identify trends in market shifts. For the remaining flags, since these observations still represent valid Eurostat records, they were retained in the dataset as is. Finally, it should be noted that the **Motor energy** attribute in this dataset does not provide information on hybrid or electric power trains. The preprocessed dataset is summarised in **Tab. 7**.

Table 7: Structure of the preprocessed Eurostat *New passenger cars by type of motor energy and engine size* dataset.

Attribute	Dtype	Value domain / range	Description
Motor energy	object	Diesel, Petrol, Total	Motor energy category.
Engine capacity of vehicle	object	From 1 400 to 1 999 cm³, 2 000 cm³ or over, Less than 1 400 cm³, Total	Engine capacity category.
geo	object		ISO2 country code of the reporting country. Restricted to EU27_2020 countries.
Geopolitical entity (reporting)	object		Description of the reporting country 'geo'.
TIME_PERIOD	int64		Year of the observation. Restricted from 2014 to 2024 .
OBS_VALUE	float64		Number of newly registered passenger cars.

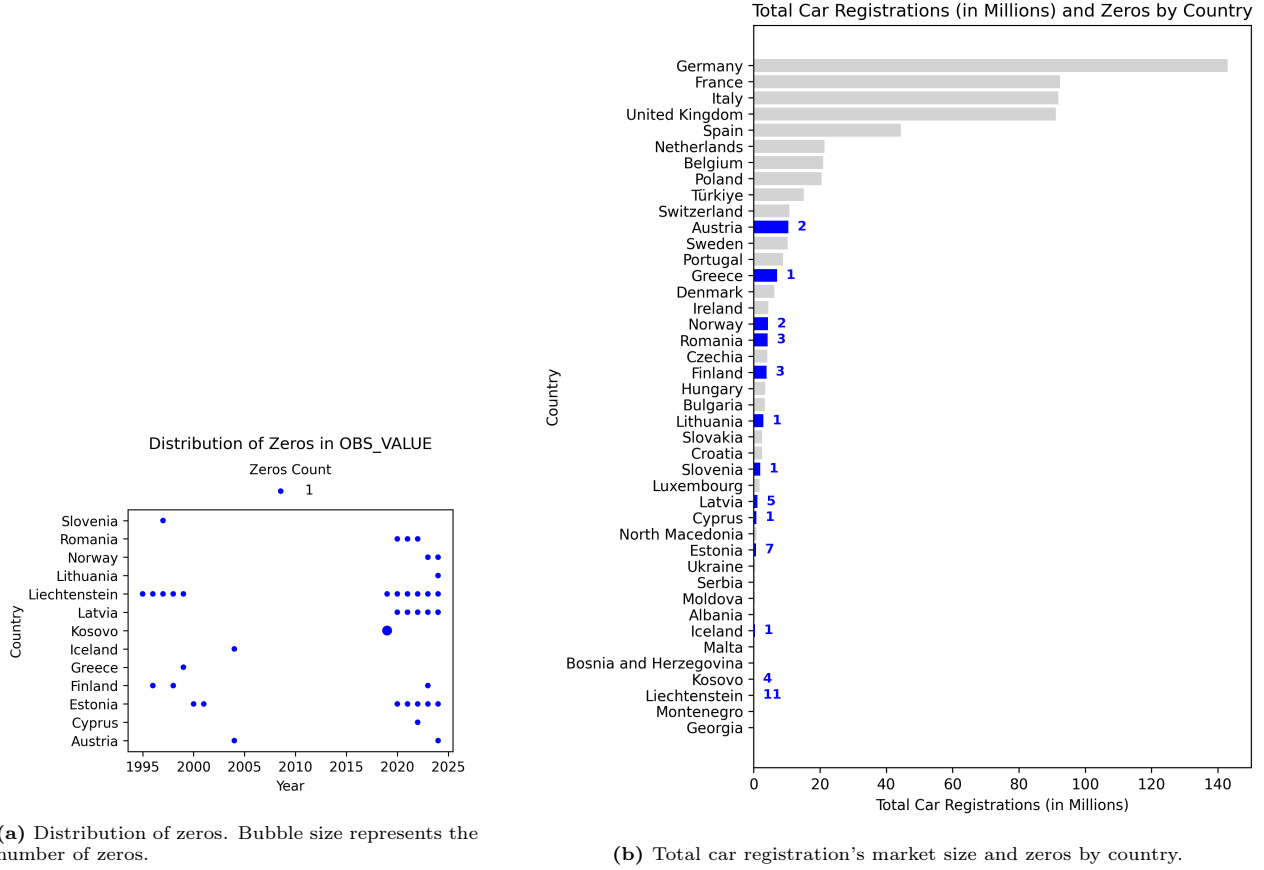


Figure 1: Eurostat *New passenger cars by type of motor energy and engine size* dataset.

3.2 Eurostat's New passenger cars by type of motor energy

The same data inspection and preprocessing strategy described for the *New passenger cars by type of motor energy and engine size* dataset was applied to this dataset.

Due to the higher granularity of the **Motor energy** attribute, this dataset naturally contains a larger number of zero-valued observations. Therefore, instead of analysing the geographical distribution of zeros, the distribution was examined across different motor energy categories instead. As shown in **Fig. 2**, most zero-valued observations are associated with more niche fuel types.

A single missing value was identified in **OBS_VALUE**, corresponding to **(Bi-fuel, Norway, 2023)** flagged with **OBS_FLAG = 'm'** (missing value). Since all other observations for **(Bi-fuel, Norway)** reported **OBS_VALUE = 0.0**, this value was substituted in place of the NaN.

Observation flags were also inspected following the same procedure adopted for the previous dataset. Although **b** (break in time series) flags are sparse, most are concentrated in **2013**. Consequently, restricting the analysis to the period **2014 onwards** still removes most affected observations.

The **Motor energy** attribute was then inspected to obtain the required granularity level and improve consistency with the other datasets. In particular, niche categories were aggregated into the *Other* category. Hybrid categories were renamed, resulting in the categories reported in **Tab. 8**.

Table 8: Structure of the preprocessed Eurostat *New passenger cars by type of motor energy* dataset.

Attribute	Dtype	Value domain / range	Description
Motor energy	object	Other, Diesel (all), Diesel (excluding hybrids), Electricity, Diesel hybrid, Diesel plug-in hybrid, Petrol hybrid, Petrol plug-in hybrid, Petrol (all), Petrol (excluding hybrids), Total	Motor energy category. Renamed and aggregated for better clarity within the scope of our analysis.
geo	object		ISO2 country code of the reporting country. Restricted to EU27_2020 countries.
Geopolitical entity (reporting)	object		Description of the reporting country 'geo'.
TIME_PERIOD	int64		Year of the observation. Restricted from 2014 to 2024 .
OBS_VALUE	float64		Number of newly registered passenger cars.

Furthermore, motor energy categories counts were inspected over time to identify possible inconsistencies in reporting practices. It should be noted that for some countries, the sum of detailed categories (e.g., *Diesel (excluding hybrids)*, *Diesel hybrid*, and *Diesel plug-in hybrid*) **before 2021** does not correspond to the reported aggregated category (e.g., *Diesel (all)*). This difference can be attributed to the voluntariness of reporting

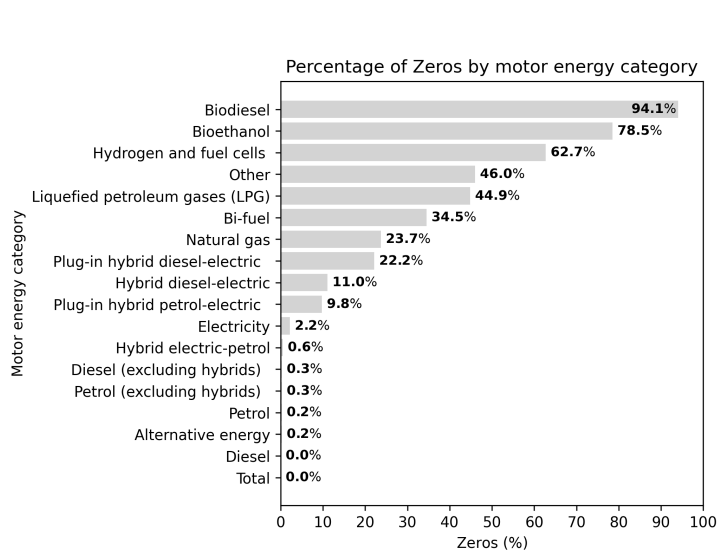


Figure 2: Percentage of zeros by motor energy category in the Eurostat *New passenger cars by type of motor energy* dataset.

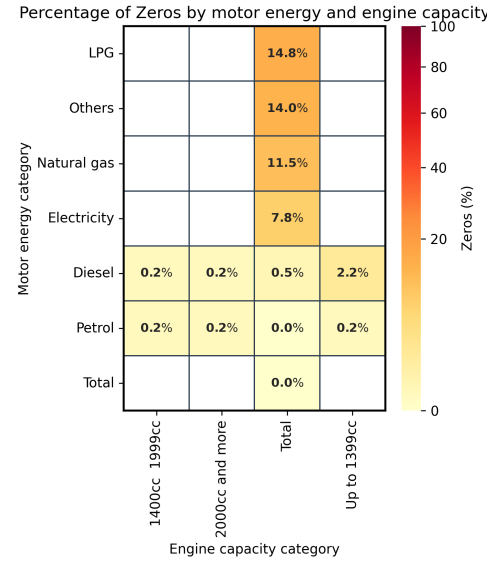


Figure 3: Percentage of zeros by motor energy category and engine capacity in the UNECE *New passenger car registrations by fuel type (detailed)* dataset.

of detailed power train registrations, prior to the introduction of harmonisation regulations across European member states.

3.3 UNECE's New passenger car registrations by fuel type (detailed)

The same data inspection and preprocessing strategy described for the Eurostat datasets was applied to the modified UNECE *New passenger car registrations by fuel type (detailed)* dataset of **Subsec. 2.3**.

An initial inspection of zero-valued observations was performed for the OBS_VALUE attribute. As shown in **Fig. 3** the distribution of zeros does not indicate a significant data quality issue, since are primarily associated with niche power train options. However, unlike the Eurostat sources, the dataset presents widely distributed missing values in OBS_VALUE across different countries, years and motor energy categories, as shown in **Fig. 4**, which cannot be reliably imputed. Furthermore, the dataset does not provide complete coverage of the **EU27_2020** member states required for this study. Therefore, the UNECE dataset was discarded from subsequent analyses due to insufficient geographical and data consistency.

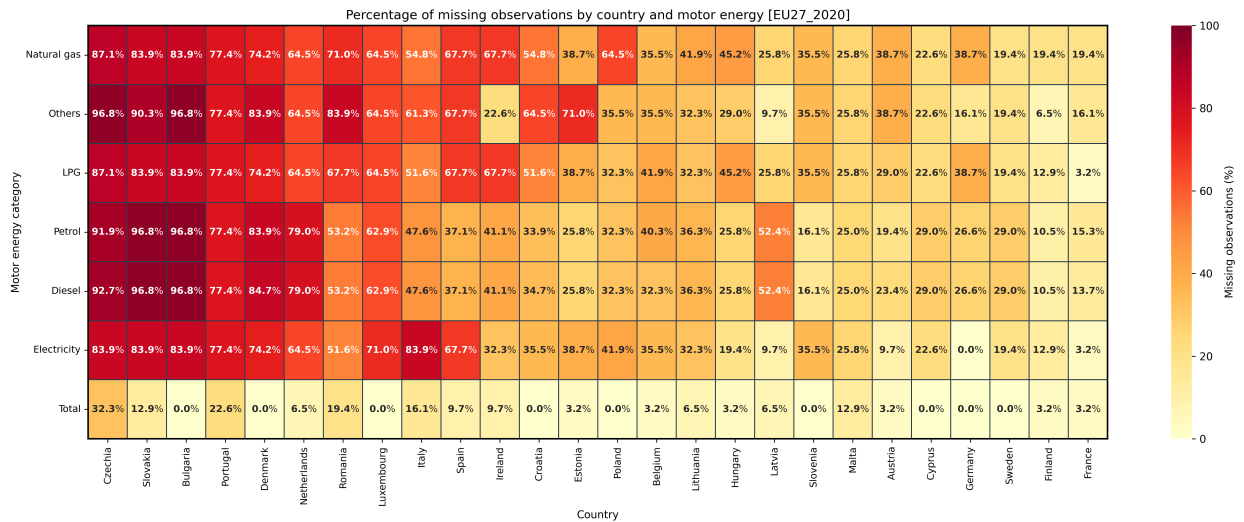


Figure 4: Distribution of missing observations across countries and motor energy categories in the UNECE *New passenger car registrations by fuel type (detailed)* dataset.

3.4 EEA’s Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631

The dataset proved challenging to analyse. It could not be loaded into memory using the pandas library, despite the raw CSV theoretically fitting in RAM, as 64GB of memory did not suffice. The dataset was therefore processed using *PySpark*, a framework for large scale data processing which does not require the entire dataset to reside in memory. Columns were remapped as shown in **Tab. 9** for ease of analysis and to align naming conventions with the Eurostat datasets. The dataset was then converted to the **Parquet** format and saved to disk. Parquet provides efficient compression and faster queries execution compared to CSV files. The resulting dataset occupies only 436MB on disk, representing a substantial reduction from the original ≈ 17 GB CSV file.

Table 9: Renamed columns of the EEA’s *Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631* dataset.

Attribute	Renamed attribute
Country	geo
VFN	vehicle_family_id_number
Mh	manufacturer_name_eu_standard_denomination
T	type
Va	variant
Ve	version
Mk	make
Cn	commercial_name
m (kg)	mass_in_running_order (kg)
Ewltpl (g/km)	co2_emissions_WLTP (g/km)
Ft	Motor energy
Fm	fuel_mode
ec (cm3)	engine_capacity (cm3)
ep (KW)	engine_power (KW)
z (Wh/km)	electric_energy_consumption (Wh/km)
r	registrations
year	TIME_PERIOD

Rows containing missing values in the **geo**, **commercial_name** and **Motor energy** attributes were removed, as these values were crucial to our analyses and could not be imputed. Approximately 180 thousand rows were removed, resulting in a dataset still containing 70M+ observations. Furthermore, superfluous attributes including **vehicle_family_id_number**, **version**, **make**, **fuel_mode** and **type** were discarded.

An inspection of the single attributes was then performed to identify inconsistencies and harmonise the values with the Eurostat dataset for the validation step. The **Motor energy** attribute was analysed first, revealing differences in capitalisation, separators and names for equivalent categories. These values were standardised and mapped into more suitable broader categories. Moreover, the *Petrol plug-in hybrid* category was ultimately merged into *Petrol hybrid*, as it was reported only for a single year and was therefore considered insufficiently represented to justify a separate analysis. Non informative categories (e.g. *unknown* or *other*) were also removed, as they accounted only for a negligible fraction of the total observations. The resulting **Motor energy** categories are reported in **Tab. 10**.

Table 10: Aggregated motor energy categories for the EEA’s *Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631* dataset.

Motor energy	Aggregate category
electric	Electricity
petrol/electric, hybrid/petrol/e, petrol phev	Petrol hybrid
diesel/electric	Diesel hybrid
petrol	Petrol (excluding hybrids)
diesel	Diesel (excluding hybrids)
biodiesel, cng, e85, gnl, hydrogen, lpg, ng, ng/biomethane, petrol/gas, other	Alternative/Other

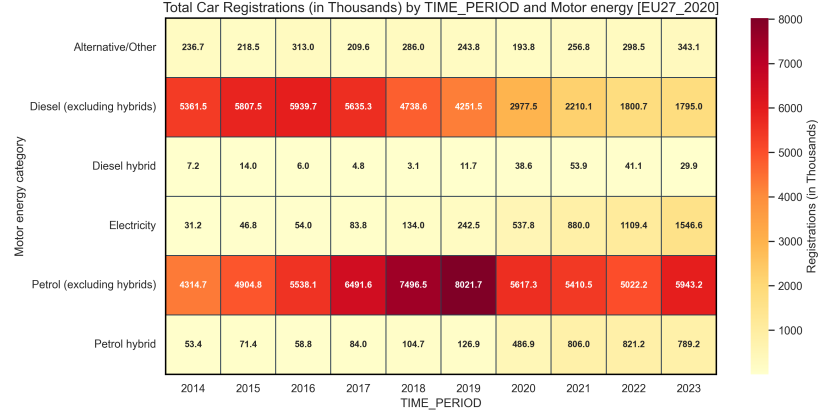
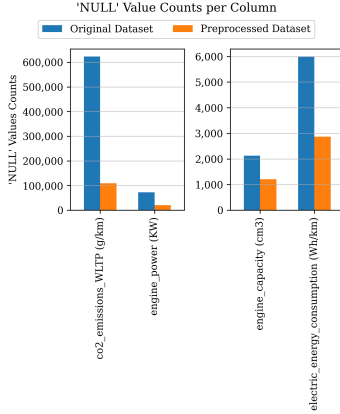
An inspection was then carried out on the vehicle specification attributes that would eventually be used for the proposed normalisations. A large number of *null* values were present in the columns **co2_emissions_WLTP** (g/km), **engine_capacity** (cm3), **engine_power** (KW) and **electric_energy_consumption** (Wh/km). A preliminary analysis suggested that most *null* values could be recovered by retrieving repeated occurrences of the same (or similar) vehicle model registered in a different country. A cleaning step was implemented, which searched for vehicles with similar characteristics, computed a distance metric for each retrieved vehicle and assigned the attribute value of the best candidate. Each distance metric is computed as the sum of the absolute differences between selected vehicle attributes. The candidate selection criteria and the distance metric adopted for each recovered attribute are described below:

- The first pass for **co2_emissions_WLTP** (g/km) involved candidates with the same **commercial_name**, **variant** and **Motor energy**, with the columns **mass_in_running_order** (kg) and **engine_capacity** (cm3) used to compute the distance metric.
- The second pass for **co2_emissions_WLTP** (g/km) involved candidates with the same **commercial_name** and **Motor energy**, computing distance as the absolute difference in **mass_in_running_order** (kg) plus the absolute differences of **engine_capacity** (cm3) and **engine_power** (KW) weighted by factors of 2 and 5, respectively.

- The pass for recovering the **engine_capacity** (cm3) attribute values involved grouping candidate vehicles with the same **commercial_name**, **variant**, and **Motor energy** to compute their rounded average engine capacity, using **mass_in_running_order** (kg) and **co2_emissions_WLTP** (g/km) to evaluate distance.
- The pass for recovering the **engine_power** (KW) attribute values involved finding candidates according to the same attributes as the previous pass, using the **mass_in_running_order** (kg) column for computing the distance metric.
- The pass for recovering the **electric_energy_consumption** (Wh/km) attribute values involved finding candidates according to the same attributes as the previous pass, using the **engine_power** (KW) and **mass_in_running_order** (kg) columns to compute the distance metric.

The recovery passes were also executed accounting for the differences in **Motor energy** category (e.g., electric vehicles were excluded from the recovery of **co2_emissions_WLTP** (g/km)). The reduction of *null* values achieved is reported in **Fig. 5a**.

After these steps, the dataset was aggregated according to the remaining attributes by summing the corresponding counts of the **registrations** attribute. Finally, the descriptive country attribute **Geopolitical entity (reporting)** was added, obtaining an efficient data representation containing approximately 2M observations, with a reported disk space usage of 20MB. The resulting overview of the evolution of registrations across different motor energy categories is presented in **Fig. 5b**.



(a) Number of *null* values before and after the cleaning step.

(b) Overview heatmap of passenger car registrations by TIME_PERIOD and Motor energy category.

Figure 5: Preprocessed EEA Monitoring of CO2 emissions from passenger cars Regulation (EU) 2019/631 dataset.

3.5 Validation of the preprocessed EEA dataset

Before proceeding with the exploratory analysis, the registration counts derived from the preprocessed EEA dataset are validated against the Eurostat *New passenger cars by type of motor energy* dataset. This dataset was selected among the complementary aggregated datasets primarily because it was the only source providing detailed registration statistics for hybrid and electrified power trains, which are essential for the objectives of this study. The temporal coverage of the EEA dataset was restricted to match the coverage period of the preprocessed Eurostat reference dataset described in **Subsec. 3.2**. The selected common period is 2014-2023.

To validate the EEA dataset, the registration counts were first aggregated by country, year and motor energy category by summing the corresponding **registrations** values. To match the Eurostat dataset, the two following aggregated motor energy categories were also computed:

$$Petrol (all) = Petrol (excluding hybrids) + Petrol hybrid$$

$$Diesel (all) = Diesel (excluding hybrids) + Diesel hybrid$$

Discrepancies were then visualised using a heatmap, where the difference between the aggregated EEA and Eurostat is expressed in thousands and computed as: $Difference (in Thousands) = X_{EEA} - X_{Eurostat} / 1000$.

As shown in **Fig. 6**, the aggregated *Petrol (all)*, *Diesel (all)*, and *Electricity* categories exhibit only negligible differences between the two datasets, validating our preprocessing steps. In contrast, discrepancies are observed for the detailed (*excluding hybrids*) and *hybrid* categories, where differences in one category are almost entirely compensated by the other. This behaviour suggests that the discrepancies do not originate from

errors in the preprocessing process, but rather from differences in the classification criteria adopted by the two data sources. Although both the Eurostat and the EEA exclude *Mild Hybrid Electric Vehicles (MHEVs)* from the hybrid categories because of their technical similarity to conventional combustion vehicles [7, 5], the Eurostat relies on national registration data whereas the EEA dataset derives the data from a more granular vehicle-by-vehicle information. Together with possible differences in reporting methodology, this provides a plausible explanation for the observed redistribution of observations.

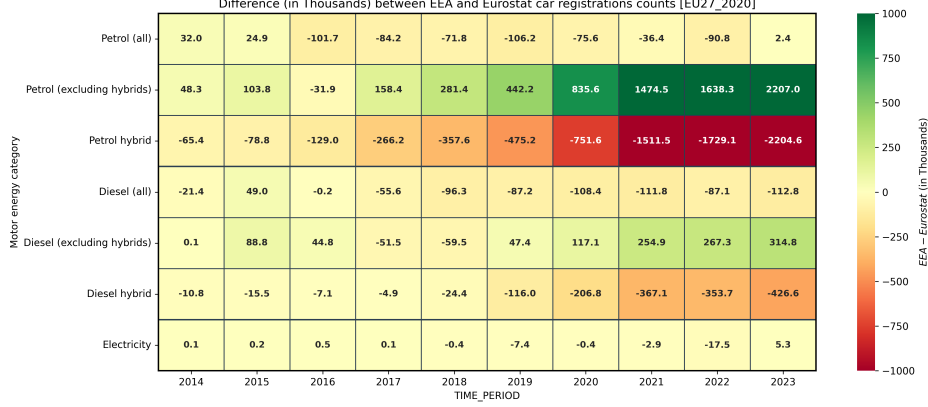


Figure 6: Absolute difference (in thousands of units) between preprocessed EEA and Eurostat passenger car registration counts.

Having established the consistency of the preprocessing steps on the EEA dataset, the remainder of this report focuses solely on this dataset, as it provides the necessary information for the proposed project.

4 Univariate, Bivariate and Multivariate analysis

Based on the univariate summaries presented in **Tab. 11, 12, 13**, several distinct patterns emerge across the dataset’s attributes.

The temporal coverage spans from 2014 to 2023, covering a decade of vehicle registrations. Across this period, the **registrations** attribute is strongly right-skewed, with a median of 3 contrasted with a mean of 50.20, thus indicating a high volume of niche vehicle configurations alongside a few popular models reaching up to 92,652 registrations.

Regarding physical specification attributes, **mass_in_running_order** (kg), **engine_capacity** (cm3) and **engine_power** (KW) show relatively symmetric distributions, with median values of 1,615 kg, 1,968 cm³ and 110 KW, respectively. Environmental metrics reveal a **co2_emissions_WLTP** (g/km) average of 151.14 g/km, closely aligned with the median, although some extreme outliers are present (max 2,810 g/km), while **electric_energy_consumption** (Wh/km) averages 195.79 Wh/km but is populated for only about 10% of the observations.

This sparsity in metrics concerning electric vehicles aligns with the categorical distribution of **Motor energy** when aggregated by registration volume. In fact, traditional ICE powertrains overwhelmingly dominate total registrations, with *Petrol* (53.34%) and *Diesel* (36.78%) accounting for over 90% of overall registrations, while pure electric vehicles account for 4.24% and various hybrid and alternative configurations make up the remainder. Finally, the dataset shows substantial market fragmentation at the model level, with 144 unique manufacturers collectively offering 31,974 distinct commercial vehicle names.

Table 11: Summary statistics for numerical attributes in the dataset.

Variable / Attribute	count	mean	stddev	min	median	max
TIME_PERIOD	2194200.00	2019.43	2.62	2014.00	2020.00	2023.00
registrations	2194200.00	50.20	368.36	1.00	3.00	92652.00
mass_in_running_order (kg)	2193332.00	1673.42	364.16	205.00	1615.00	5440.00
co2_emissions_WLTP (g/km)	2077594.00	151.14	59.16	0.00	151.00	2810.00
engine_capacity (cm3)	2103584.00	1901.76	697.85	120.00	1968.00	8382.00
engine_power (KW)	2173610.00	136.07	71.92	0.00	110.00	1770.00
electric_energy_consumption (Wh/km)	215437.00	195.79	64.18	1.00	182.00	1754.00

Table 12: Unique counts for high-cardinality categorical attributes.

manufacturer_unique_count	commercial_name_unique_count
144	31974

To examine the interaction between physical vehicle dimensions and environmental performance, **Fig. 7** illustrates the weighted average WLTP CO₂ emissions as a function of **mass_in_running_order** (kg) across

Table 13: Distribution of newly registered passenger cars by motor energy category.

Motor energy	registrations	percentage (%)
Petrol (excluding hybrids)	58760697	53.34
Diesel (excluding hybrids)	40517447	36.78
Electricity	4666088	4.24
Petrol hybrid	3402513	3.09
Alternative/Other	2599858	2.36
Diesel hybrid	210233	0.19

different **Motor energy** categories. For conventional internal combustion engines, a strong positive linear relationship is evident. *Petrol (excluding hybrids)* exhibits the steepest emissions trajectory, ascending from approximately 105 g/km at 850 kg to a peak near 350 g/km at $2,750 \text{ kg}$. *Diesel (excluding hybrids)* follows a parallel upward trend but maintains a consistently lower baseline across equivalent weight classes, climbing from 90 g/km at $1,050 \text{ kg}$ to 290 g/km at $2,800 \text{ kg}$, demonstrating diesel’s higher efficiency in moving heavier vehicle frames.

In contrast, hybrid power trains display a different pattern. *Petrol hybrid* emissions remain stable at $110\text{--}120 \text{ g/km}$ below $1,500 \text{ kg}$ (largely driven by non-plug-in HEVs), before dropping to $30\text{--}40 \text{ g/km}$ past $1,500 \text{ kg}$, as battery assistance decouples vehicle mass from tailpipe emissions. A similar emission profile characterizes *Diesel hybrid* vehicles, which appear almost exclusively in heavier weight classes ($> 1,900 \text{ kg}$) and average under 50 g/km across most mass bins.

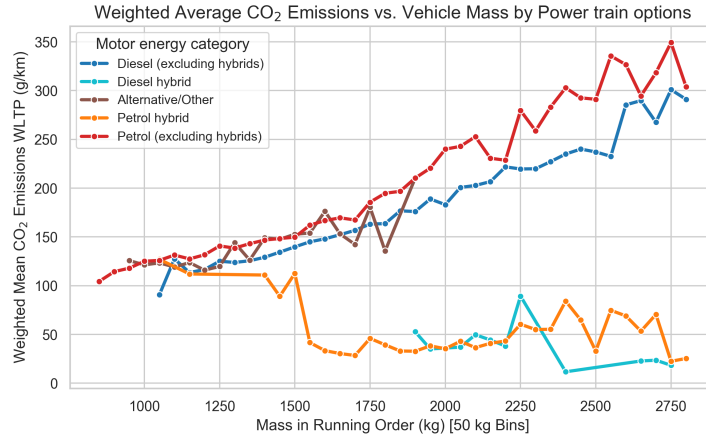


Figure 7: Weighted average WLTP CO₂ emissions (g/km) versus vehicle mass in running order (kg) binned in 50 kg increments, categorized by motor energy category for the preprocessed EEA dataset.

To evaluate structural weight differences across power train technologies, **Fig. 8** depicts the relative registration distribution across **mass_in_running_order** (kg) for each **Motor energy** category. Conventional combustion engines show a clear mass separation. *Petrol (excluding hybrids)* is predominantly concentrated in lighter weight classes (peaking at 12.5% around $1,150 \text{ kg}$) due to its prevalence in compact urban passenger car models. Conversely, *Diesel (excluding hybrids)* displays a broader and heavier distribution centered between $1,350 \text{ kg}$ and $1,650 \text{ kg}$, with a maximum of 10.5% around $1,500 \text{ kg}$. *Alternative/Other* fuels show a distribution more similar to petrol, with most registrations concentrated between $1,150 \text{ kg}$ and $1,350 \text{ kg}$ and peaks reaching 15.3% .

In contrast, electrified and hybrid power trains demonstrate substantial weight inflation driven by battery integration and platform sizing. *Petrol hybrid* registrations center in mid-to-large vehicle segments, forming a distinct peak of 13.1% at $1,800 \text{ kg}$. Pure *Electricity* exhibits a broader multi-modal distribution spanning from $1,400 \text{ kg}$ to $2,200 \text{ kg}$, featuring a prominent spike at $1,600 \text{ kg}$ (13.4%) alongside secondary clusters near $1,800 \text{ kg}$ and $2,050 \text{ kg}$. *Diesel hybrid* represents the heaviest category, occurring almost exclusively above $1,900 \text{ kg}$ with concentration peaks at $2,150 \text{ kg}$ (22.3%) and $2,650 \text{ kg}$ (17.1%), reflecting its niche deployment in heavy luxury SUVs and executive platforms.

To evaluate how vehicle mass impacts electrical efficiency across electrified power trains, **Fig. 9** illustrates **electric_energy_consumption** (Wh/km) as a function of **mass_in_running_order** (kg) for the EV and PHEV subset. For pure battery electric vehicles (*Electricity*), energy consumption remains relatively stable across light-to-mid weight classes, ranging between 140 Wh/km and 170 Wh/km from $1,000 \text{ kg}$ to $2,000 \text{ kg}$. Above $2,000 \text{ kg}$, consumption follows a clear upward trend, reaching peaks of $260\text{--}265 \text{ Wh/km}$ around $2,650\text{--}2,750 \text{ kg}$. This demonstrates that while battery electric power trains absorb vehicle mass penalties efficiently at lower weights, heavy luxury and SUV platforms incur substantial energy overheads.

Plug-in hybrid configurations show broader consumption variance and steeper growth at higher mass thresholds. *Petrol hybrid* models first appear around $1,400 \text{ kg}$, with consumption reaching a minimum of approx-

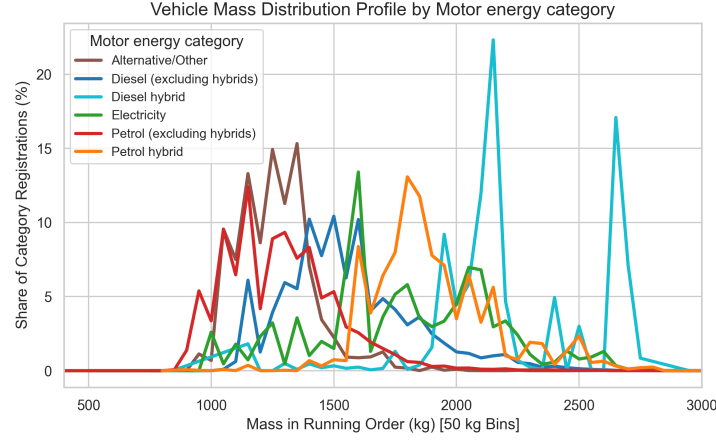


Figure 8: Normalized vehicle mass distribution profile by motor energy category (50 kg bins) for the preprocessed EEA dataset.

imately 76 Wh/km at 1,500 kg, before increasing to levels comparable with pure EVs around 1,700 kg (168 Wh/km). Above 2,400 kg, Petrol hybrid consumption exceeds pure electric values, rising to 280-296 Wh/km at the highest mass range (2,650-2,800 kg). Similarly, Diesel hybrid power trains are deployed in heavy weight class vehicles ($\geq 1,900$ kg) and show large variations, reaching peaks of 260 Wh/km at 2,100 kg and 311 Wh/km at 2,650 kg. Overall, the higher electrical demand observed for heavy PHEVs reflects the greater energy required from their electric motors and batteries to move heavier vehicles without internal combustion assistance.

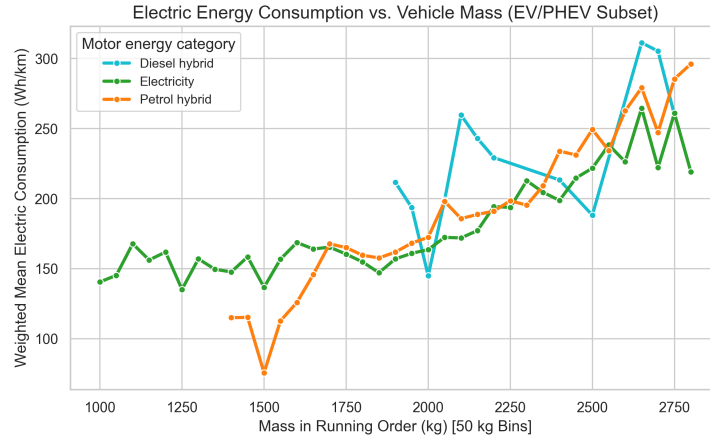


Figure 9: Weighted mean electric energy consumption (Wh/km) versus vehicle mass in running order (kg) binned in 50 kg increments for EV and PHEV powertrains for the preprocessed EEA dataset.

To evaluate multivariate interactions among numerical attributes, **Fig. 10** presents the Pearson correlation matrix across physical, mechanical, and environmental metrics. A strong triad emerges between `mass_in_running_order` (kg), `engine_capacity` (cm3) and `engine_power` (KW). The strongest correlation is between engine capacity and engine power ($r = 0.85$), while vehicle mass is also positively correlated with engine capacity ($r = 0.66$) and engine power ($r = 0.57$). In this three way interaction, greater vehicle mass is associated with larger vehicle structures, which in turn tend to require larger engines and higher power outputs to maintain comparable vehicle performance. When considering also the environmental metrics, engine capacity serves as the primary mechanical driver of `co2_emissions_WLTP` (g/km) ($r = 0.55$), whereas engine power demonstrates a weaker relationship ($r = 0.34$), suggesting that engine capacity has an higher physical influence on fuel consumption and emissions rather than peak horsepower output.

A temporal dynamic captured in **Fig. 10** is the policy driven decoupling between vehicle weight growth and carbon emissions over time. Between 2014 and 2023, `TIME_PERIOD` is positively correlated with `mass_in_running_order` (kg) ($r = 0.21$), confirming that newly registered passenger cars became progressively heavier over the decade due to SUV market expansion and battery integration. At the same time, `TIME_PERIOD` exhibits a negative correlation with `co2_emissions_WLTP` (g/km) ($r = -0.20$), suggesting a gradual decoupling between vehicle mass and CO₂ emissions over the analysed period, potentially reflecting regulatory pressure and improvements in vehicle efficiency. Furthermore, this temporal shift may help explain why the overall lin-

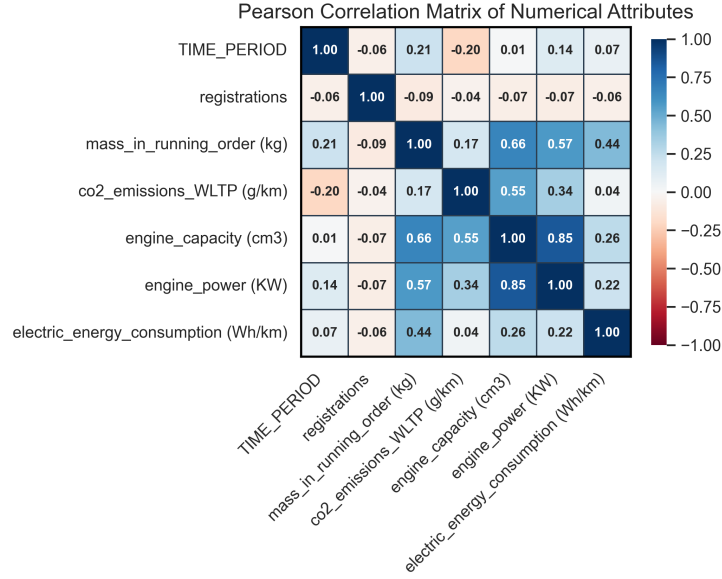


Figure 10: Pearson correlation matrix across numerical vehicle attributes for the preprocessed EEA dataset.

ear correlation between `mass_in_running_order` (kg) and `co2_emissions_WLTP` (g/km) appears relatively weak ($r = 0.17$) across the unsegmented dataset. As heavier electrified power trains entered the market at scale, their low or zero tailpipe emissions offset the historical positive relationship between vehicle mass and CO₂ emissions.

5 Time series analysis

The temporal evolution of passenger car registrations was analysed using the preprocessed EEA dataset to investigate how the European car market for conventional internal combustion engine (ICE) and electrified power trains has changed over time. Since market shares better encode the transition between different power train technologies by accounting for variations in the total market size, the analysis considered both the number of registrations and the relative market share of each motor energy category over the available period (2014-2023).

Furthermore, to estimate an evolution of the observed trends, a forecast has been applied up to 2028 (year chosen given the limited observation period available) as follows:

- The annual number of registrations was calculated, obtaining the market volume that was then extended up to 2028 using *Holt Winter's Exponential Smoothing* method, as it usually yields low sMAPE (symmetric Mean Absolute Percentage Error) and sMedAPE (symmetric Median Absolute Percentage Error) values even for short time series [3]. The additive (constant) trend component (`trend="add"`) was selected given the limited variations in historical data.
- The market share of each motor energy category was then computed and extended to 2028 using the same approach but with a multiplicative trend component (`trend="mul"`) to better capture the adoption process. The predicted shares were subsequently normalized to ensure a total sum equal to 100%.
- Finally, the estimated market shares for each year and motor energy category were converted into registration counts by multiplying for their respective market volume estimation.

As shown in **Fig. 11**, the estimated trends suggest a continued decline in *Petrol* and, especially, *Diesel* vehicles, accompanied by an increase in *Electrified* options, reflecting the ongoing transition towards a more sustainable passenger car market.

6 Feature selection

As introduced in the project proposal, two complementary normalization approaches are considered to account for both the number and characteristics of the vehicle options available to consumers, supporting a more representative visualization of the choices available when purchasing a new passenger car:

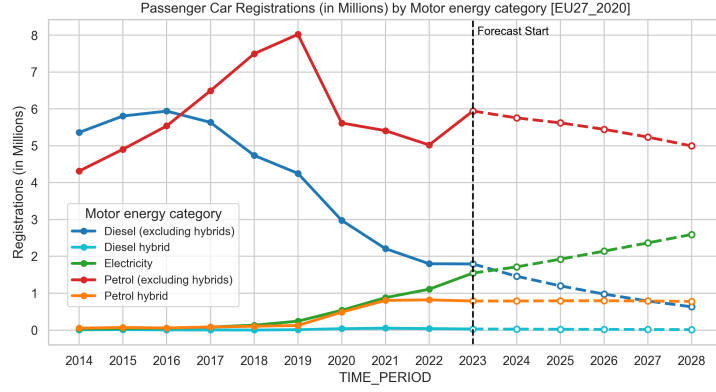


Figure 11: Evolution of passenger car registrations by motor energy category for the preprocessed EEA dataset.

- **Normalisation by Attributes (Baseline).** As a baseline, the number of available vehicle options is quantified for each country, year and **Motor energy** category by counting the unique combinations of **commercial_name**, **engine_capacity** (cm3) and **engine_power** (KW). This provides a discrete measure of the technical configurations available to consumers while both considering vehicle design and limiting repeated observations of the same configuration and excessive granularity that could arise from including attributes such as **variant** or **co2_emissions_WLTP** (g/km), which may distinguish minor equipment or configuration differences rather than distinct vehicle offerings.
- **Autoencoder-Based Normalisation.** The second approach represents vehicle options according to their technical characteristics rather than their identifiers. Five continuous numerical attributes are selected to train the autoencoder: **mass_in_running_order** (kg), **engine_power** (KW), **engine_capacity** (cm3), **co2_emissions_WLTP** (g/km) and **electric_energy_consumption** (Wh/km). Together, these attributes describe the physical, mechanical, and environmental characteristics of each vehicle. Metadata (**geo**, **TIME_PERIOD** and **registrations**), text identifiers (**manufacturer_name_eu_standard_denomination**, **commercial_name** and **variant**) and **Motor energy** are excluded from the input features. In particular, excluding **Motor energy** allows vehicles from different power train categories to be represented within a common latent space based strictly on their physical and functional utility.

The primary motivation for training the network on these features is to enable non-linear dimensionality reduction that overcomes the limitations of discrete, text-based model counting techniques. In fact, counting unique vehicle names or trim combinations can artificially inflate market availability due to homologation redundancies and platform sharing. Instead, by compressing the selected physical specifications into a lower-dimensional (2D or 3D) continuous latent space, the autoencoder groups vehicles according to their functional characteristics. The geometric volume or spatial occupancy of each power train type can then be estimated through grid binning, providing a normalized measure of market diversity and the breadth of choice available to consumers across time.

References

- [1] Ankur Agarwal, Shashi Prabha, and Raghav Yadav. Exploratory data analysis (eda) for banking and finance: Unveiling insights and patterns. *arXiv preprint arXiv:2407.11976*, 2024. URL: <https://arxiv.org/pdf/2407.11976>.
- [2] Ahmad Alshammari. Implementation of exploratory data analysis (eda) in python. *International Journal of Computer Applications*, 187:34–42, 08 2025. doi:10.5120/ijca2025925577.
- [3] Ruben Crevits and Christophe Croux. Forecasting using robust exponential smoothing with damped trend and seasonal components. *SSRN*, 2017. doi:10.2139/ssrn.3068634.
- [4] Data-Centric AI Community. fg-data-profiling: 1 line of code data quality profiling & exploratory data analysis for Pandas and Spark DataFrames. <https://github.com/Data-Centric-AI-Community/fg-data-profiling>, 2026.
- [5] European Environment Agency. Emep/eea air pollutant emission inventory guidebook 2023: 1.a.3.b.i-iv road transport. Technical report, European Environment Agency, 2025. URL: <https://copert.emisia.com/wp-content/uploads/2025/10/1.A.3.b.i-iv-Road-transport-2025.pdf>.
- [6] Eurostat. Member States EU27 (from 2020) trade by BEC product group. URL: https://ec.europa.eu/eurostat/databrowser/view/ext_st_27_2020msbec/default/table, doi:10.2908/ext_st_27_2020msbec.
- [7] Eurostat. Guidance note - reporting of electric and more resource-efficient transport equipment in EPEA and EGSS accounts. Technical report, European Environmental Economic Accounts, 2020. URL: <https://ec.europa.eu/eurostat/documents/1798247/12177560/Guidance+note+on+electric+transport+equipment+-+technical+note.pdf/2ddec6dc-8ca9-1736-0f36-18ed2233af0b?t=1609859296315>.